THE COLOURED SAND THEOREM: THE p-PART OF THE SANDPILE GROUP OF THE HAMMING GRAPH

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Abstract. Let $H(d,q) = K_q^d$ be the Hamming graph and $K(H(d,q))$ its sandpile group (critical group). For primes $p \nmid q$ its p-part was determined by Jacobson, Niedermaier and Reiner in 2003, who called the remaining primes difficult, "due to bad behavior of its p-primary structure". We treat the primes $p \mid q$. Jacobson, Niedermaier and Reiner showed that over $\mathbb{Z}$ the Laplacian splits into explicit triangular pieces; we read these pieces as operators $q(c+H)+U$ on the Boolean lattice (A). From this we prove: (B) a closed formula for the Sylow p-subgroup for every prime $p > d$ and every q divisible by p, in which ballot numbers appear; (C) the exponent of the Sylow p-subgroup for every prime p and every $q \geq 3$, namely $p^{\max_{1\leq s\leq d}(es+v_p(s))}$ with $e = v_p(q) \geq 1$; (D) the number of its cyclic factors when $p \mid q$, from Wilson's diagonal form of inclusion matrices. For every prime p and every $d \leq p^2+p-1$ we determine the group exactly (F, G), by writing down the lifts to the p-adic integers of the tilting modules of SL_2 in characteristic p that have two Weyl factors. For $p = 2$ we determine it for every d (H), from the results of [DST] on the hypercube. This proves the Glue Conjecture of version 1 and Conjecture H of version 2; for these results 107 predictions were written down before they were computed (44 in version 1, 63 in version 3), all correct. We give a closed form for the number of cyclic factors when $3 \mid q$, and we conjecture the shape of the remaining blocks (I): the number of large cyclic factors is the number of Weyl factors of the tilting module, a power of 2 read off the p-adic digits, and the small ones do not depend on q.

1. Introduction

For a connected graph Γ with Laplacian $L(\Gamma)$, the sandpile group $K(\Gamma)$ is the torsion subgroup of $\operatorname{coker} L(\Gamma)$, and $\operatorname{coker} L(\Gamma) \cong \mathbb{Z} \oplus K(\Gamma)$; its order is the number of spanning trees [Big99]. The Hamming graph $H(d,q)$ has vertex set $(\mathbb{Z}/q)^d$, two vertices being adjacent when they differ in exactly one coordinate; it is the Cartesian power K_q^d of the complete graph. Its Laplacian eigenvalues are qs, $0 \leq s \leq d$, with multiplicity $\binom{d}{s}(q-1)^s$.

Jacobson, Niedermaier and Reiner [JNR03] determined the Sylow p-subgroup of $K(K_{n_1} \times \cdots \times K_{n_k})$ for every prime p that divides none of the n_i [JNR03, Theorem 2], and bounded the number of invariant factors [JNR03, Theorem 3]. About the other primes they wrote: "Describing the exact structure of the critical group $K(K_{n_1} \times \cdots \times K_{n_k})$ appears to be difficult, due to bad behavior of its p-primary structure for primes p that divide some of $n_1, \ldots, n_k$." Ducey and Jalil [DJ14] recovered the results of [JNR03] by another method. For $q = 2$ the graph is the hypercube; its odd part is due to Bai [Bai03] and its 2-part was determined in [DST].

In this paper $q \geq 2$ is arbitrary, p is a prime dividing q, and $e := v_p(q) \geq 1$, except where we say otherwise.

The pieces. Let B_m be the free abelian group on the subsets of $[m] = \{1, \ldots, m\}$, graded by size: $B_m = \bigoplus_{s=0}^{m} B_m(s)$. Let H act on $B_m(s)$ as the scalar s, and let U and D be the up and down operators,

$$U(S) = \sum_{x \notin S} S \cup \{x\}, \qquad D(S) = \sum_{x \in S} S \setminus \{x\}.$$

For integers $c \geq 0$ put

$$X_q(m,c) := \operatorname{coker}\big(q(c+H)+U \ \text{ on } B_m\big).$$

It is finite when $c \geq 1$; for $c = 0$ it is $\mathbb{Z}$ plus a finite group.

Date: 13 October 2026 (version 3).

Theorem A (Jacobson–Niedermaier–Reiner [JNR03, Theorem 13])**.** *For every $q \geq 2$ and $d \geq 1$, as abelian groups,*

$$\operatorname{coker} L\big(H(d,q)\big) \;\cong\; \bigoplus_{j=0}^{d} X_q(d-j,\, j)^{\oplus \binom{d}{j}(q-2)^j}.$$

This is the case $n_1 = \cdots = n_d = q$ of [JNR03, Theorem 13]: there the pieces are upper triangular matrices $M(N; n_S)$ of size $2^{|S|}$ with identity blocks above the diagonal, and after reversing the order of the subsets of S they are the matrices $q(c + H) + U$. We include the short proof below, in the form we use. For $q = 2$ only $j = 0$ survives and the theorem says $\operatorname{coker} L(Q_d) \cong X_2(d, 0)$, the starting point of [DST].

Theorem B (the primes $p > d$)**.** *Let $d \geq 1$, let $p > d$ be a prime dividing q, and $e = v_p(q)$. Then*

$$\operatorname{Syl}_p K\big(H(d,q)\big) \;\cong\; \bigoplus_{v=1}^{d} \big(\mathbb{Z}/p^{ev}\big)^{n_v}, \qquad n_v = \sum_{\substack{j+2k=d+1-v \\ (j,k)\neq(0,0)}} \binom{d}{j} (q-2)^j\, b(k,v),$$

the sum over integers $j, k \geq 0$, where $b(k,v) = \frac{v}{v+2k}\binom{v+2k}{k} = [x^k]\, C(x)^v$ are the ballot numbers and $C(x) = \sum_n \frac{1}{n+1}\binom{2n}{n} x^n$ is the Catalan series.

Equivalently, with $c(z) = C(z^2)$ and $u = q - 2$, $\sum_v n_v y^v = [z^{d+1}]\,(1+uz)^d\, \frac{yz\,c(z)}{1-yz\,c(z)} - y^{d+1}$. For instance, with $u = q - 2$:

$$\begin{aligned}
d = 2: &\quad (\mathbb{Z}/p^e)^{u^2+1} \oplus (\mathbb{Z}/p^{2e})^{2u},\\
d = 3: &\quad (\mathbb{Z}/p^e)^{u^3+3u} \oplus (\mathbb{Z}/p^{2e})^{3u^2+2} \oplus (\mathbb{Z}/p^{3e})^{3u},\\
d = 4: &\quad (\mathbb{Z}/p^e)^{u^4+6u^2+2} \oplus (\mathbb{Z}/p^{2e})^{4u^3+8u} \oplus (\mathbb{Z}/p^{3e})^{6u^2+3} \oplus (\mathbb{Z}/p^{4e})^{4u}.
\end{aligned}$$

Theorem B is the case $m < p$ of a statement about single pieces (Theorem E in Section 3), which also gives, for every prime p, the contribution of all pieces with $d - j < p$.

Theorem C (the exponent)**.** *Let $q \geq 3$, let p be any prime and $e = v_p(q)$. The exponent of $\operatorname{Syl}_p K(H(d,q))$ is p^E, where*

$$E = \max_{1\leq s\leq d} \big(e\,s + v_p(s)\big) \quad \text{if } e \geq 1, \qquad E = \lfloor \log_p d \rfloor \quad \text{if } e = 0.$$

For $p > d$ this gives $E = ed$, as in Theorem B. For $p \leq d$ the maximum need not be at $s = d$: for $q = 3$ and $d = 10$ it is at $s = 9$ and $E = 11$. The case $e = 0$ ($p \nmid q$) can also be read off from [JNR03, Theorem 2]; we include it because the proof below covers it at no cost.

Theorem D (the number of cyclic factors)**.** *Let $p \mid q$. The number of cyclic factors of $\operatorname{Syl}_p K(H(d,q))$ is*

$$N = \sum_{j=0}^{d} \binom{d}{j} (q-2)^j \Big(2^{\,d-j} - \rho_p(d-j)\Big) \;-\; 1, \qquad \rho_p(m) = \sum_{s=0}^{m-1} r_p(m,s),$$

where $r_p(m,s) = \sum_i \big(\binom{m}{i} - \binom{m}{i-1}\big)$, the sum over $0 \leq i \leq t$ with $p \nmid k - i$, and $(t,k) = (s, s+1)$ if $2s \leq m-1$, $(t,k) = (m-1-s, m-s)$ otherwise.

Here $\rho_p(m)$ is the rank over $\mathbb{F}_p$ of U on B_m. For $q = p = 3$ the theorem gives $N = 1, 4, 12, 34, 96, 274, \ldots$, and $N = 3^{d-1} + 2^{d-1} - 1$ for every d (Corollary 6). For $q = p$ the value of N equals the bound $\frac{1}{2}(q^d + (q-2)^d) - 1$ of [JNR03, Theorem 3] for $d \leq 2$, is one less for $d = 3$, and is strictly smaller for $4 \leq d \leq 8$ and $p \leq 11$ (where we compared).

Corollary 1. *$\operatorname{Syl}_p K(H(d,q))$ depends on q only through $e = v_p(q)$ and the multiplicities $\binom{d}{j}(q-2)^j$.*

Status. This is a preprint. The proofs were re-read by the system that wrote them (see Provenance) and checked against the computations of Section 9; they have not yet been read by an independent reader. The versions are on Zenodo, concept doi:10.5281/zenodo.23327608 (version 1: doi:10.5281/zenodo.23327609; version 2: doi:10.5281/zenodo.23328702). Version 2 (13 October 2026) added Theorems F and G for $d \le 3p-1$ and Corollary 6; in version 1, Theorem F was stated as a conjecture. Version 3 (13 October 2026) extends Theorem G to $d \le p^2+p-1$ and every prime, which contains Conjecture H of version 2, adds Theorem H (the prime 2, every d) and the new Conjecture I.

By Theorems G and H, what remains is p odd and $d \ge p^2+p$, and there only the pieces $X_q(m,c)$ with $m \ge p^2+p$, whose tilting summands can have four or more Weyl factors. For these blocks we state Conjecture I (Section 8).

2. The splitting (after Jacobson, Niedermaier and Reiner)

Proof of Theorem A. Let $K_q = qI - J$ be the Laplacian of the complete graph K_q, acting on $\mathbb{Z}^q$ with basis $e_0,\dots,e_{q-1}$, and put $f_i = e_i - e_0$ $(1 \le i \le q-1)$ and $\sigma = f_1 + \cdots + f_{q-1}$. Then $e_0, \sigma, f_1, \dots, f_{q-2}$ is a basis of $\mathbb{Z}^q$ (unimodular change from $e_0, f_1, \dots, f_{q-1}$), and

$$K_q e_0 = -\sigma, \qquad K_q \sigma = q\sigma, \qquad K_q f_i = q f_i.$$

So $\mathbb{Z}^q = A \oplus W$ with $A = \mathbb{Z}e_0 \oplus \mathbb{Z}\sigma$ and $W = \bigoplus_{i \le q-2} \mathbb{Z} f_i$, both stable under K_q, and K_q is the scalar q on W. The Laplacian of $H(d,q)$ is $L = \sum_{i=1}^{d} 1 \otimes \cdots \otimes K_q \otimes \cdots \otimes 1$ on $(\mathbb{Z}^q)^{\otimes d} = \bigoplus_{T \subseteq [d]} A^{\otimes([d]\setminus T)} \otimes W^{\otimes T}$, and each summand is L-stable. On the summand of T with $|T| = j$, L acts as $(L_A + qj) \otimes 1$, where $L_A = \sum_i K_q^{(i)}$ on $A^{\otimes(d-j)}$ and $W^{\otimes T} \cong \mathbb{Z}^{(q-2)^j}$. There are $\binom{d}{j}$ sets T of size j.

Index the basis of $A^{\otimes m}$ by the subsets $S \subseteq [m]$, the factor in position i being σ if $i \in S$ and e_0 otherwise. Then $L_A(S) = q|S|\,S - \sum_{x \notin S} S \cup \{x\}$, that is, $L_A = qH - U$, and $L_A + qj = q(j+H) - U$. Conjugating by the diagonal matrix $(-1)^{|S|}$ turns $-U$ into U. Hence the summand contributes $\operatorname{coker}(q(j+H)+U \text{ on } B_{d-j}) = X_q(d-j,j)$, $\binom{d}{j}(q-2)^j$ times. □

Proof of Corollary 1. By Theorem A it suffices to see that $\operatorname{Syl}_p X_q(m,c)$ depends only on the p-adic valuations of the diagonal entries $a_s = q(c+s)$. This is Lemma 2. □

Lemma 2 (only the valuations matter)**.** *Let $a_0,\dots,a_m$ be elements of $\mathbb{Z}_{(p)}$ and $M = \operatorname{diag}(a_{|S|}) + U$ on $B_m \otimes \mathbb{Z}_{(p)}$. If $a_s = p^{w_s}u_s$ with units u_s (or $a_s = 0$, with $u_s = 1$), then M is equivalent over $\mathbb{Z}_{(p)}$ to*

$$\operatorname{diag}(p^{w_{|S|}}) + U \qquad (\textit{with } p^w := 0 \textit{ when } a_s = 0).$$

Proof. Look for units λ_s, μ_s such that $P = \operatorname{diag}(\lambda_{|S|})$ and $Q = \operatorname{diag}(\mu_{|S|})$ give $PMQ = \operatorname{diag}(p^{w_{|S|}}) + U$. The diagonal of PMQ on rank s is $\lambda_s a_s \mu_s$ and its block from rank s to rank $s+1$ is $\lambda_{s+1}\mu_s U$, so we need $\lambda_s \mu_s u_s = 1$ and $\lambda_{s+1}\mu_s = 1$. Put $\mu_0 = 1$, $\lambda_0 = u_0^{-1}$, and recursively $\lambda_{s+1} = \mu_s^{-1}$, $\mu_{s+1} = (\lambda_{s+1}u_{s+1})^{-1}$. □

3. Strings, and the primes $p > m$

On B_m one has $[D,U] = m - 2H$: on a set S of size s, $DU(S) = (m-s)S + \Sigma$ and $UD(S) = s\,S + \Sigma$ with the same sum $\Sigma = \sum_{y \in S,\, x \notin S} (S \setminus \{y\}) \cup \{x\}$. For $0 \le k \le m/2$ let $\Lambda_k = \ker D \cap B_m(k)$. By induction on i, for $v \in \Lambda_k$,

$$D\,U^i v = i\,(m - 2k - i + 1)\,U^{i-1}v. \tag{1}$$

Over $\mathbb{Q}$ this is the classical $\mathfrak{sl}_2$-structure of the Boolean lattice [Pro82]:

$$B_m \otimes \mathbb{Q} = \bigoplus_{k \le m/2} \bigoplus_{i=0}^{m-2k} U^i \Lambda_k^{\mathbb{Q}},$$

$\dim \Lambda_k^{\mathbb{Q}} = \binom{m}{k} - \binom{m}{k-1}$, and $U^{m-2k+1}\Lambda_k = 0$. We call $\Lambda_k, U\Lambda_k, \dots, U^{m-2k}\Lambda_k$ a *string*; its ranks are $k, \dots, m-k$ and its length is $m - 2k + 1$.

Lemma 3 (strings over $\mathbb{Z}_{(p)}$)**.** *If $p > m$, then $B_m \otimes \mathbb{Z}_{(p)} = \bigoplus_{k\le m/2} \bigoplus_{i=0}^{m-2k} U^i\Lambda_k$, each Λ_k is free of rank $\binom{m}{k} - \binom{m}{k-1}$, and $U^i\colon \Lambda_k \to U^i\Lambda_k$ is an isomorphism for $i \le m-2k$.*

Proof. All lattices below are over $R = \mathbb{Z}_{(p)}$. First, for $r \le m/2$ we show $B_m(r) = UB_m(r-1) \oplus \Lambda_r$, by induction on r (for $r = 0$, $\Lambda_0 = B_m(0)$). By induction $B_m(r-1) = \bigoplus_{k<r} U^{r-1-k}\Lambda_k$, and by (1) the operator DU acts on $U^{r-1-k}\Lambda_k$ as the scalar $(r-k)(m-k-r+1)$; both factors lie in $[1, m]$, so they are units. Hence DU is invertible on $B_m(r-1)$, $\pi = U(DU)^{-1}D$ is an idempotent on $B_m(r)$ with image $UB_m(r-1)$, and its kernel is $\ker D \cap B_m(r) = \Lambda_r$ because $U(DU)^{-1}$ is injective. This gives the decomposition, and Λ_r is a direct summand, so free, of rank $\binom{m}{r} - \binom{m}{r-1}$.

Now fix any rank r and consider $\Phi\colon \bigoplus_{k\le \min(r,m-r)} \Lambda_k \to B_m(r)$, $(v_k) \mapsto \sum_k U^{r-k}v_k$. For $k' \le r$ let $\psi_{k'}\colon B_m(r) \to \Lambda_{k'}$ be $D^{r-k'}$ followed by the projection of $B_m(k') = UB_m(k'-1) \oplus \Lambda_{k'}$ onto $\Lambda_{k'}$. By (1), $D^{r-k'}U^{r-k}v_k$ is 0 if $k > k'$ (since $Dv_k = 0$), lies in $U^{k'-k}\Lambda_k \subseteq UB_m(k'-1)$ if $k < k'$, and equals $\prod_{i=1}^{r-k} i(m-2k-i+1)\, v_k$ if $k = k'$, a unit multiple of v_k (all factors lie in $[1, m]$ because $k \le m-r$). So $(\psi_{k'}) \circ \Phi$ is triangular with unit diagonal, Φ is split injective, and its image is a direct summand of $B_m(r)$ of full rank (compare ranks over $\mathbb{Q}$). Hence Φ is an isomorphism. Finally $U^{m-2k+1}\Lambda_k = 0$ holds over $\mathbb{Q}$, hence over R. □

Each string is stable under $q(c+H)+U$: in the basis $v, Uv, \dots, U^{\ell-1}v$ ($\ell = m-2k+1$) the operator is lower bidiagonal with diagonal entries $q(c+k+i)$, $0 \le i < \ell$, and all subdiagonal entries 1. The cokernel of such a matrix over a principal ideal domain is cyclic of order its determinant, because the minor obtained by deleting the first row and last column equals 1; if one diagonal entry is 0 (only possible for $c = k = i = 0$) the cokernel is $\mathbb{Z}_{(p)}$. This proves:

Theorem E (pieces with $m < p$)**.** *Let p be any prime, $e = v_p(q)$, and $m < p$. Then*

$$\operatorname{Syl}_p X_q(m,c) \;\cong \bigoplus_{\substack{0\le k\le m/2\\ (c,k)\neq(0,0)}} \big(\mathbb{Z}/p^{A_k}\big)^{\binom{m}{k}-\binom{m}{k-1}}, \qquad A_k = \sum_{r=k}^{m-k} \big(e + v_p(c+r)\big),$$

and for $c = 0$ the string $k = 0$ contributes only the free summand $\mathbb{Z}$.

Proof of Theorem B. If $p > d$, every piece $X_q(d-j, j)$ has $m = d-j < p$ and $c+r \le d < p$, so $v_p(c+r) = 0$ for $c+r \ge 1$ and $A_k = e(m-2k+1)$. Put $v = m-2k+1 = d+1-j-2k$; then $\binom{m}{k} - \binom{m}{k-1} = \binom{v+2k-1}{k} - \binom{v+2k-1}{k-1} = \frac{v}{v+2k}\binom{v+2k}{k} = b(k,v)$. Summing over j with Theorem A gives the formula. The generating function follows from $\sum_k b(k,v)x^k = C(x)^v$. □

4. The exponent

Lemma 4. *Let $c \ge 1$ and $M = q(c+H)+U$ on B_m. Then M is invertible over $\mathbb{Q}$, and for $S \subseteq T$ with $|S| = s$, $|T| = s+k$,*

$$(M^{-1})_{T,S} = (-1)^k\, k! \prod_{i=0}^{k} \frac{1}{q(c+s+i)},$$

while $(M^{-1})_{T,S} = 0$ if $S \not\subseteq T$.

Proof. With $\Delta = q(c+H)$, $M^{-1} = (1+\Delta^{-1}U)^{-1}\Delta^{-1} = \sum_{k\ge0}(-\Delta^{-1}U)^k\Delta^{-1}$, a finite sum. A path from S to T in $(\Delta^{-1}U)^k\Delta^{-1}$ is a maximal chain $S = S_0 \subset \cdots \subset S_k = T$, there are $k!$ of them, and each collects Δ^{-1} once at every rank from s to $s+k$. □

For a nonsingular square matrix over a discrete valuation ring with uniformizer p, the exponent of the cokernel is p^ϵ with $-\epsilon$ the least valuation of an entry of M^{-1} (write M in Smith form). Hence, for $c \ge 1$, the exponent of $\operatorname{Syl}_p X_q(m,c)$ is p to the power

$$\max_{\substack{s,k\ge0\\ s+k\le m}} \Big(\sum_{i=0}^{k} \big(e + v_p(c+s+i)\big) - v_p(k!)\Big) = \max_{c\le a\le b\le c+m} \Big(eL + v_p(L) + v_p\tbinom{b}{L}\Big), \quad L = b-a+1,$$

where $a = c + s$, $b = c + s + k$, and we used $\frac{b!}{(a-1)!\,(b-a)!} = L\binom{b}{L}$.

Lemma 5 (carries)**.** *For $d \geq 1$ and $e \geq 1$,*

$$\max_{1 \leq L \leq b \leq d} \Big(eL + v_p(L) + v_p\tbinom{b}{L} \Big) = \max_{1 \leq s \leq d} \big(es + v_p(s) \big),$$

and for $e = 0$ the left side equals $\lfloor \log_p d \rfloor$.

Proof. Taking $L = b = s$ gives $\geq$. Let $\lambda = \lfloor \log_p b \rfloor$. By Kummer's theorem $v_p\binom{b}{L}$ is the number of carries in the addition $L + (b - L)$ in base p. No carry occurs at a position below $v_p(L)$, where the digits of L vanish, and a carry out of position i forces $b \geq p^{i+1}$, so $i \leq \lambda - 1$. Hence $v_p(L) + v_p\binom{b}{L} \leq \lambda$. For $e = 0$ this gives the bound $\lfloor \log_p d \rfloor$, attained at $L = b = p^{\lfloor \log_p d \rfloor}$.

Let $e \geq 1$; we induct on b, the case $b = 1$ being clear. Let $s^* = p^\lambda \lfloor b/p^\lambda \rfloor$, so $1 \leq s^* \leq b < s^* + p^\lambda$ and $v_p(s^*) \geq \lambda$. If $L \leq s^*$, then $eL + v_p(L) + v_p\binom{b}{L} \leq es^* + \lambda \leq es^* + v_p(s^*)$. If $L > s^*$, write $b = s^* + r$ and $L = s^* + r'$ with $1 \leq r' \leq r < p^\lambda$. Then $v_p(L) = v_p(r')$, and since $(b - L) + r' = r < p^\lambda$, the additions $L + (b - L)$ and $r' + (r - r')$ have the same carries, so $v_p\binom{b}{L} = v_p\binom{r}{r'}$. By induction there is $1 \leq t \leq r$ with $er' + v_p(r') + v_p\binom{r}{r'} \leq et + v_p(t)$. Since $t < p^\lambda \leq p^{v_p(s^*)}$, $v_p(s^* + t) = v_p(t)$, and so $eL + v_p(L) + v_p\binom{b}{L} \leq e(s^* + t) + v_p(s^* + t)$ with $s^* + t \leq b$. □

Proof of Theorem C. By Theorem A the exponent is the largest exponent of the pieces $\mathrm{Syl}_p X_q(d - j, j)$ that occur. For $j \geq 1$ the piece occurs ($q \geq 3$) and the computation above gives the maximum over $j \leq a \leq b \leq d$; for $j = 1$ this is the full range $1 \leq a \leq b \leq d$. For $j = 0$, the row of the empty set in $M_0 = qH + U$ is zero and M_0 maps into $B' = \bigoplus_{s \geq 1} B_d(s)$, so

$$\operatorname{coker} M_0 = \mathbb{Z} \oplus B'/(M'' B' + \mathbb{Z}u),$$

where M'' is the restriction of M_0 to B' (nonsingular, lower triangular by rank with diagonal qs) and $u = M_0(\emptyset)$. Its torsion is a quotient of $\operatorname{coker} M''$, whose exponent is given by Lemma 4 (with $c = 0$ and $s \geq 1$) as the same maximum over $1 \leq a \leq b \leq d$. So the exponent is p to $\max_{1 \leq a \leq b \leq d}(eL + v_p(L) + v_p\binom{b}{L})$, and Lemma 5 finishes. □

For $q = 2$ the same argument gives only an upper bound, and it is not attained: for Q_2 the group is $\mathbb{Z}/4$ while the bound is 2^3.

5. The number of cyclic factors

Proof of Theorem D. The number of cyclic factors of $\mathrm{Syl}_p K$ is $\dim_{\mathbb{F}_p}(\operatorname{coker} L \otimes \mathbb{F}_p) - 1$, the 1 for the free summand. Modulo p, $q(j + H) + U \equiv U$, so by Theorem A $\dim_{\mathbb{F}_p}(\operatorname{coker} L \otimes \mathbb{F}_p) = \sum_j \binom{d}{j}(q-2)^j \big(2^{d-j} - \operatorname{rank}_{\mathbb{F}_p} U|_{B_{d-j}}\big)$. The rank of U is the sum over s of the ranks of the inclusion matrices $W_{s,s+1}$ of s-subsets versus $(s+1)$-subsets of $[m]$. By Wilson's diagonal form [Wil90], for $t \leq \min(k, m-k)$ the matrix $W_{t,k}$ is equivalent over $\mathbb{Z}$ to a diagonal matrix with entries $\binom{k-i}{t-i}$ of multiplicity $\binom{m}{i} - \binom{m}{i-1}$, $0 \leq i \leq t$ (we use the statement as given in [CSX15, (2)]); so its rank over $\mathbb{F}_p$ is $\sum_{i:\, p \nmid \binom{k-i}{t-i}} \big(\binom{m}{i} - \binom{m}{i-1}\big)$. For $W_{s,s+1}$ with $2s \leq m - 1$ take $(t, k) = (s, s+1)$, so $\binom{k-i}{t-i} = s + 1 - i$; otherwise complementation identifies $W_{s,s+1}$ with the transpose of $W_{m-s-1,m-s}$ and we take $(t, k) = (m - 1 - s, m - s)$. This is $r_p(m, s)$. □

Corollary 6 (the number of factors when $3 \mid q$)**.** *If $3 \mid q$, the number of cyclic factors of $\mathrm{Syl}_3 K(H(d, q))$ is*

$$N = (q-1)^d + \frac{q^d - (q-1)^d}{3} - \frac{(q-1)^d - (q-3)^d}{6} - 1.$$

In particular $N = 3^{d-1} + 2^{d-1} - 1$ for $q = 3$.

Proof. By Theorem D it suffices to know $f(m) = \dim_{\mathbb{F}_3} \operatorname{coker}(U|_{B_m \otimes \mathbb{F}_3})$. Identify $B_m \otimes \mathbb{F}_3$ with $V^{\otimes m}$, $V = \mathbb{F}_3^2$, so that $U = \sum_i e_i$ with e the nilpotent 2×2 Jordan block acting on the i-th factor. The e_i commute, so $U^3 = \sum_i e_i^3 = 0$, and $V^{\otimes m}$ is a module over the Hopf algebra $\mathbb{F}_3[e]/(e^3)$ with e primitive. Then $f(m)$ is the number of Jordan blocks of U. Write J_a for the indecomposable module of dimension $a \leq 3$. Then:

- $J_1 \otimes J_2 = J_2$;
- $J_2 \otimes J_2 = J_3 \oplus J_1$, since $e^2(v_0 \otimes v_0) = 2\, v_1 \otimes v_1 \neq 0$;
- $J_3 \otimes J_2 = J_3^2$, because J_3 is free, a projective tensored with any module is projective, and projectives over this local algebra are free.

So $V^{\otimes m} = a_m J_1 \oplus b_m J_2 \oplus c_m J_3$ with $(a_{m+1}, b_{m+1}, c_{m+1}) = (b_m, a_m, b_m + 2c_m)$, starting from $(1, 0, 0)$. Hence $a_m + b_m = 1$, $b_m = [m \text{ odd}]$, and counting dimensions, $c_m = (2^m - 1 - [m \text{ odd}])/3$. This gives $f(m) = 1 + (2^m - 1 - [m \text{ odd}])/3$. Summing $\sum_m \binom{d}{m}(q-2)^{d-m} f(m)$ gives the formula. □

6. The range $p \leq d$: tilting lattices

By Theorem E the only pieces not yet described are $X_q(m, c)$ with $m \geq p$. For $p \leq m$ the strings of length $> p$ break: by (1) the coefficient $i(m - 2k - i + 1)$ vanishes modulo p. The way they break is governed by the representation theory of SL_2 in characteristic p.

The dictionary. Let $V = \mathbb{Z}_p v_0 \oplus \mathbb{Z}_p v_1$, where $\mathbb{Z}_p$ is the ring of p-adic integers, and let f act on V by $f v_0 = v_1$, $f v_1 = 0$; give v_0 weight 1 and v_1 weight -1. Identify $B_m \otimes \mathbb{Z}_p$ with $V^{\otimes m}$ by $S \mapsto v_{\mathbf{1}_S}$. Then U is the action of f on the tensor power, a subset of size s has weight $m - 2s$, and so $H = (m - h)/2$ with h the weight. The divided powers $U^{(r)}$ (add r new elements, in all ways) are integral, and so is the analogous operator $e^{(r)}$ (remove r elements). Hence $B_m \otimes \mathbb{Z}_p$ is a module over the distribution algebra $\operatorname{Dist}(SL_2)_{\mathbb{Z}_p}$, and

$$q(c + H) + U \;=\; q\big(c + \tfrac{m-h}{2}\big) + f$$

is built from that action alone.

We use three standard facts.

(T1) $V^{\otimes m}$ is a tilting module, and it decomposes over $\mathbb{Z}_p$ as $\bigoplus_\lambda T_{\mathbb{Z}_p}(\lambda)^{n_\lambda}$, where $T_{\mathbb{Z}_p}(\lambda)$ is the unique lift of the indecomposable tilting module $T(\lambda)$ over $\mathbb{F}_p$. The reason is that $\operatorname{Ext}^1(\Delta, \nabla) = 0$, so decompositions over $\mathbb{F}_p$ lift; this is standard [Jan03, Don93]. For $p = 2$ the argument is written out, and checked in Lean, in [DST, Lemma 3.3, Corollary 3.4].

(T2) Let $\lambda \leq p^2 + p - 1$. If $\lambda \leq p - 1$, or if $\lambda + 1 \equiv 0 \pmod p$ and $\lambda \leq p^2 - 1$, then $T(\lambda) = \Delta(\lambda)$ is simple. Otherwise write

$$\lambda = p - 1 + r + pa, \qquad 1 \leq r \leq p,\ 0 \leq a \leq p - 1,$$

where $r = p$ (and then $a = p - 1$) occurs only for $\lambda = p^2 + p - 1$. Then $T(\lambda)$ has exactly two Weyl factors, $\Delta(\lambda)$ and $\Delta(\lambda^-)$ with $\lambda^- := \lambda - 2r$, and it is a non-split extension $0 \to \Delta(\lambda) \to T(\lambda) \to \Delta(\lambda^-) \to 0$; so is its lift over $\mathbb{Z}_p$. Indeed, by Donkin's tensor product theorem [Don93], $T(p - 1 + r' + pb) \cong T(p - 1 + r') \otimes T(b)^{[1]}$ for $0 \leq r' \leq p - 1$. For $r \leq p - 1$ take $r' = r$, $b = a \leq p - 1$, so that $T(a) = \Delta(a)$. For $\lambda = p^2 + p - 1$ take $r' = 0$, $b = p$: then $T(p - 1)$ is the Steinberg module and $T(p)$ has the Weyl factors $\Delta(p), \Delta(p - 2)$. In both cases the product of characters, expanded by Brauer's formula with $\chi(-n - 2) = -\chi(n)$, leaves exactly $\chi(\lambda) + \chi(\lambda - 2r)$; the other terms cancel in pairs.

(T3) Comparing characters, the multiplicities in $V^{\otimes m}$ are given, from the top down, by $n_\lambda = \beta(m, \lambda) - n_{\lambda^+}$. Here λ^+ is the weight whose bottom factor is λ, and $n_{\lambda^+} = 0$ if there is none or if $\lambda^+ > m$. (For $\lambda \leq p^2 + p - 1$ the map $\lambda \mapsto \lambda^-$ is injective: $\lambda^- \equiv p - 1 - r \pmod p$ determines r, and then a.)

For $\lambda \equiv m \pmod 2$, $0 \leq \lambda \leq m$, write $\beta(m, \lambda) = \binom{m}{(m-\lambda)/2} - \binom{m}{(m-\lambda)/2 - 1}$ (the number of strings of length $\lambda + 1$).

The blocks. Fix m, c and $\lambda \equiv m \pmod 2$, put $k = (m-\lambda)/2$ and

$$v_t = e + v_p(c+k+t) \qquad (0 \le t \le \lambda),$$

with $v_0 = \infty$ when $c = k = 0$. Put $A = \sum_{t=0}^{\lambda} v_t$. When $T(\lambda)$ has two Weyl factors, with (r, a) as in (T2), put also

$$A' = \sum_{t=r}^{r+\lambda^-} v_t, \qquad \alpha = \sum_{t=(a+1)p}^{\lambda} v_t, \qquad \tau = 1 + [a = p-1],$$

and

$$E_{11} = A - \alpha - a, \qquad E_{12} = A' - a + \tau, \qquad E_{22} = A' - a + \alpha.$$

Here A' is the sum over the levels of the bottom factor (it is $A_c(m, \lambda^-)$ in the notation of Theorem F), and α is the sum over the last r levels of the top factor. Define the abelian p-group $G_c(m,\lambda)$ by

$$G_c(m,\lambda) = \begin{cases} \mathbb{Z}/p^A, & \lambda \le p-1, \\ (\mathbb{Z}/p)^a \oplus \mathbb{Z}/p^{A-a}, & \lambda + 1 = (a+1)p \le p^2, \\ (\mathbb{Z}/p)^{2a} \oplus \mathbb{Z}/p^x \oplus \mathbb{Z}/p^y, & \text{two Weyl factors}, \end{cases}$$

where $x = \min(E_{11}, E_{12}, E_{22})$ and $y = E_{11} + E_{22} - x$. When $c = k = 0$ (the block containing the empty set) we put instead: $\mathbb{Z}_p$ for $\lambda \le p-1$; $\mathbb{Z}_p \oplus (\mathbb{Z}/p)^a$ in the second case; and $\mathbb{Z}_p \oplus (\mathbb{Z}/p)^{2a} \oplus \mathbb{Z}/p^{\min(E_{12},E_{22})}$ in the third.

When $a + 1 < p$ one has $\tau = 1$ and $x = E_{12} = A' + 1 - a$, so the third case reads $(\mathbb{Z}/p)^{2a} \oplus \mathbb{Z}/p^{A'+1-a} \oplus \mathbb{Z}/p^{A-1-a}$, which is the block of version 2. When $a = p-1$ all three numbers E_{ij} can be the minimum: for $p = 3$, $\lambda = 10$, $k = 0$, $e = 1$ the pairs (x, y) are $(11, 11)$ for $c = 1$ and $(10, 12)$ for $c = 2$.

The first theorem is the statement conjectured in version 1 of this paper. It is the case $m \le 2p-2$ of Theorem G below, where only $a = 0$ occurs.

Theorem F (the Glue Theorem; Conjecture F of version 1)**.** *Let p be a prime, q an integer with $e = v_p(q) \ge 1$, $m \le 2p-2$ and $c \ge 0$. Then*

$$\begin{aligned} \operatorname{Syl}_p X_q(m,c) \;\cong\; & \bigoplus_{\lambda \le p-1} \big(\mathbb{Z}/p^{A_c(m,\lambda)}\big)^{\beta(m,\lambda) - \beta'(m,\lambda)} \\ & \oplus \bigoplus_{p \le \lambda \le m} \Big(\mathbb{Z}/p^{A_c(m,\lambda)-1} \oplus \mathbb{Z}/p^{A_c(m,2p-2-\lambda)+1}\Big)^{\beta(m,\lambda)}, \end{aligned}$$

where $A_c(m,\lambda) = \sum_{r=(m-\lambda)/2}^{(m+\lambda)/2} \big(e + v_p(c+r)\big)$, $\beta'(m,\lambda) = \beta(m, 2p-2-\lambda)$ if $\lambda \le p-2$ and $\beta'(m, p-1) = 0$, and λ runs over integers $\equiv m \pmod 2$. When $c = 0$ the string of length $m+1$ (through the empty set) has a zero on its diagonal: for $\lambda = m < p$ it gives nothing, and for $\lambda = m \ge p$ the term $\mathbb{Z}/p^{A_c(m,m)-1}$ is omitted (it is the free summand) while its partner $\mathbb{Z}/p^{A_0(m,2p-2-m)+1}$ stays.

More generally:

Theorem G (two Weyl factors)**.** *Let p be a prime, q an integer with $e = v_p(q) \ge 1$, $c \ge 0$, and $m \le p^2 + p - 1$. Then*

$$\mathbb{Z}_p^{[c=0]} \oplus \operatorname{Syl}_p X_q(m,c) \;\cong\; \bigoplus_{\lambda} G_c(m,\lambda)^{\,n_\lambda},$$

where λ runs over the integers $\equiv m \pmod 2$ with $0 \le \lambda \le m$ and n_λ is as in (T3).

With Theorem A, Theorem G determines $\operatorname{Syl}_p K(H(d,q))$ for every $d \le p^2 + p - 1$. This contains Conjecture H of version 2 (p odd, $d \le p^2 - 2$). For $p = 3$ the range is $d \le 11$ and for $p = 5$ it is $d \le 29$. For $p = 2$ it is $d \le 5$; the prime 2 is treated for every d in Section 7. The range is exactly the range in which every tilting summand of $V^{\otimes m}$ has at most two Weyl

factors; at $m = p^2 + p$ the block $\lambda = p^2 + p$, with four Weyl factors, appears. The hypothesis $e \geq 1$ is needed: for $p \nmid q$ the formula fails, for instance when $(q, p, d) = (4, 3, 4)$; there the answer of [JNR03] applies instead.

Proofs. Fix λ with two Weyl factors, (r, a) as in (T2), $\mu = \lambda^-$. Let $\ell_i = f^{(i)}x$ $(0 \leq i \leq \lambda)$ be the standard basis of $\Delta_{\mathbb{Z}_p}(\lambda)$, so $f\ell_i = (i+1)\ell_{i+1}$, $e\ell_i = (\lambda - i + 1)\ell_{i-1}$, and more generally $f^{(t)}\ell_i = \binom{i+t}{t}\ell_{i+t}$ and $e^{(t)}\ell_i = \binom{\lambda-i+t}{t}\ell_{i-t}$. Let $s_j = f^{(j)}y$ $(0 \leq j \leq \mu)$ be that of $\Delta_{\mathbb{Z}_p}(\mu)$. Put

$$\nu = \min_{1 \leq t \leq r} v_p\tbinom{\lambda - r + t}{t}.$$

Lemma 7 (the glue is forced)**.** *$\nu = 2$ if $r \leq p-1$ and $a = p-1$, and $\nu = 1$ otherwise. $T_{\mathbb{Z}_p}(\lambda)$ is isomorphic to the lattice in $\Delta_{\mathbb{Q}_p}(\lambda) \oplus \Delta_{\mathbb{Q}_p}(\mu)$ spanned by the ℓ_i and by*

$$\tilde{s}_j = s_j + p^{-\nu}\tbinom{j+r}{j}\,\ell_{j+r} \qquad (0 \leq j \leq \mu).$$

Proof. The value of ν follows from Kummer's theorem. If $r \leq p-1$, then $\lambda - r = (a+1)p - 1$, and adding $t \in [1, r]$ to it carries once out of the units digit; it carries a second time exactly when $a + 1 = p$. If $r = p$, then $\lambda - r = p^2 - 1$: adding $t < p$ carries twice, while adding $t = p$ carries once.

By (T2), $T = T_{\mathbb{Z}_p}(\lambda)$ contains $\Delta_{\mathbb{Z}_p}(\lambda)$ with quotient $\Delta_{\mathbb{Z}_p}(\mu)$. Since $T \cap \Delta_{\mathbb{Q}_p}(\lambda) = \Delta_{\mathbb{Z}_p}(\lambda)$ and the weight spaces of $\Delta(\lambda)$ are one-dimensional, the lifts of the s_j to T can be chosen of the form $\tilde{s}_j = s_j + x_j\ell_{j+r}$ with $x_j \in \mathbb{Q}_p$, unique modulo $\mathbb{Z}_p$.

Stability under f gives $(j + r + 1)x_j - (j+1)x_{j+1} \in \mathbb{Z}_p$. Stability under $f^{(p)}$ at $j = kp$, for $(k+1)p \leq \mu$, gives

$$\tbinom{(k+1)p+r}{p}x_{kp} - \tbinom{(k+1)p}{p}x_{(k+1)p} \in \mathbb{Z}_p.$$

In the range of (T2), $\mu \leq p^2 - 2$, so $k + 1 \leq p - 1$ and $\binom{(k+1)p}{p} \equiv k + 1$ is a unit; so is $\binom{(k+1)p+r}{p}$, which is $\equiv k + 1$ if $r \leq p - 1$ and $\equiv k + 2$ if $r = p$ (and then $\mu = p^2 - p - 1$, so $k + 2 \leq p - 1$). Inside a segment $[kp, (k+1)p - 1]$ the factor $j + 1$ is a unit, so $x_j \equiv \binom{j+r}{j}\binom{kp+r}{kp}^{-1}x_{kp}$ modulo $\mathbb{Z}_p$. Across segments, $x_{(k+1)p} \equiv \binom{(k+1)p+r}{p}\binom{(k+1)p}{p}^{-1}x_{kp}$, and

$$\tbinom{n+r}{p}\tbinom{n}{p}^{-1} = \tbinom{n+r}{n}\tbinom{n-p+r}{n-p}^{-1} \qquad (n = (k+1)p)$$

is an identity of rational numbers (both sides equal $\frac{(n+r)!\,(n-p)!}{n!\,(n-p+r)!}$). Hence $x_j \equiv \binom{j+r}{j}x_0$ modulo $\mathbb{Z}_p$ for every j, and the lattice is determined by x_0.

Stability under $e^{(t)}$ at $j = 0$ gives $x_0\binom{\lambda-r+t}{t}\ell_{r-t} \in T$ for $1 \leq t \leq r$, so $v_p(x_0) \geq -\nu$. Suppose $v_p(x_0) \geq 1 - \nu$. Then $e^{(t)}\tilde{s}_0 \in pT$ for every $t \geq 1$ (also for $t > r$, where it vanishes), so the image of $\tilde{s}_0$ in $T \otimes \mathbb{F}_p$ is a highest weight vector of weight μ. It generates the image of a map $\Delta(\mu) \to T \otimes \mathbb{F}_p$, and the composite with $T \otimes \mathbb{F}_p \to \Delta(\mu)$ is the identity on the generator, so it is an isomorphism and the extension splits. This is impossible, because $T \otimes \mathbb{F}_p = T(\lambda)$ is indecomposable. So $x_0 = u\,p^{-\nu}$ with u a unit, and rescaling y makes $u = 1$. The lattice so described is T itself, so it is stable under all $f^{(t)}$ and $e^{(t)}$; nothing more has to be checked. (For $f^{(t)}$ this is also visible directly: $f^{(t)}\tilde{s}_j - \binom{j+t}{t}\tilde{s}_{j+t}$ vanishes by the identity $\binom{j+r}{j}\binom{j+r+t}{t} = \binom{j+t}{t}\binom{j+t+r}{j+t}$.) Stability under every $e^{(t)}$ and $f^{(t)}$ is also checked by machine in Section 9. □

Lemma 8 (the blocks)**.** *Embed $T_{\mathbb{Z}_p}(\lambda)$ in $B_m \otimes \mathbb{Z}_p$ at levels $k, \dots, k + \lambda$ $(m = \lambda + 2k)$. Then the cokernel of $q(c + H) + U$ on it is $G_c(m, \lambda)$. The same holds for the simple blocks.*

Proof. Put $N_t = q(c + k + t)$, so $v_p(N_t) = v_t$, and $M = q(c + H) + U$. In the basis of Lemma 7:

- $M\ell_i = N_i\ell_i + (i+1)\ell_{i+1}$;
- $M\tilde{s}_j = N_{j+r}\tilde{s}_j + (j+1)\tilde{s}_{j+1}$ for $j < \mu$; the ℓ-component is $p^{-\nu}[(j + r + 1)\binom{j+r}{j} - (j + 1)\binom{j+r+1}{j+1}]\ell_{j+r+1} = 0$;
- $M\tilde{s}_\mu = N_{\mu+r}\tilde{s}_\mu + u\,\ell_{\mu+r+1}$, where $\mu + r + 1 = (a+1)p$ and $u = p^{-\nu}\binom{\mu+r}{\mu}(a+1)p$.

By Lucas, $\binom{\mu+r}{\mu}$ is a unit: $\mu+r=(a+1)p-1$ has digits $(a,p-1)$, or $(p-1,p-1)$ when $r=p$. So $v_p(u)=v_p((a+1)p)-\nu$. This is 0 when $r\le p-1$, and 1 when $r=p$.

The non-unit subdiagonal entries are:

- on the ℓ-chain, the $i+1=tp$ with $1\le t\le a+1$, of valuation 1 except the last, of valuation τ;
- on the $\tilde{s}$-chain, the $j+1=tp\le\mu$, of valuation 1. There are a of them if $r\le p-1$, and $a-1$ if $r=p$.

Solving the unit subdiagonals, the cokernel is generated by the first vector of each segment between them, with one relation at the end of each segment. Write $g_t=\ell_{tp}$ and $h_t=\tilde{s}_{tp}$, and let α_t, σ_t be the sums of the v's over the segments, so that $\sum_t\alpha_t=A$, $\sum_t\sigma_t=A'$ and the last α_t is α. After rescaling the generators by units, the relations are:

- $p^{\alpha_t}g_t+p\,g_{t+1}$ on the ℓ-chain, the last link having p^τ instead of p;
- $p^{\alpha}g_{a+1}$ at the end of the ℓ-chain;
- $p^{\sigma_t}h_t+p\,h_{t+1}$ on the $\tilde{s}$-chain;
- at the end of the $\tilde{s}$-chain, $p^{\sigma}h_{\text{last}}+p^{v_p(u)}g_{a+1}$.

Every entry has valuation ≥ 1, so the relation matrix is pM' with M' integral.

If $r\le p-1$ the last relation has a unit coefficient on g_{a+1}. Eliminating g_{a+1} turns the last link of the ℓ-chain into $p^{\tau+\sigma_a}h_a$ and the end relation into $p^{\alpha+\sigma_a}h_a$. In M' the links with coefficient 1 form a units on the ℓ-chain and a on the $\tilde{s}$-chain, in distinct rows and columns. Clearing them (column operations along each chain, then row operations) splits off I_{2a}. The 2×2 matrix that remains, on the rows of the last link of the ℓ-chain and of the end relation $p^{\alpha+\sigma_a}h_a$, and on the columns g_0,h_0, is upper triangular with diagonal valuations $E_{11}-1$, $E_{22}-1$ and corner $E_{12}-1$.

If $r=p$ the $\tilde{s}$-chain has $a-1$ unit links and its end relation has a unit coefficient on $g_{a+1}=g_p$ in M'. These give $2a$ units again, and clearing them leaves the same 2×2 matrix.

So the relation matrix is equivalent to $pI_{2a}\oplus\begin{pmatrix}p^{E_{11}} & *\,p^{E_{12}}\\ 0 & p^{E_{22}}\end{pmatrix}$, with a unit $*$. The Smith form of such a matrix is $(p^x,p^{E_{11}+E_{22}-x})$ with x the least of the three valuations, and this is $G_c(m,\lambda)$.

When $c=k=0$, $N_0=0$: the column g_0 is zero and gives the free summand, and what remains on h_0 are the relations $p^{E_{12}}$ and $p^{E_{22}}$. The simple blocks are single chains. When $\lambda+1=(a+1)p\le p^2$ the chain has a non-unit subdiagonal entries, $i+1=tp$ with $1\le t\le a\le p-1$, each of valuation 1. The same elimination gives $(\mathbb{Z}/p)^a\oplus\mathbb{Z}/p^{A-a}$.

When $a+1<p$ we have $E_{12}\le E_{22}$ (since $\alpha\ge 1$) and $E_{12}\le E_{11}$, because $A-\alpha=A'+\sum_{t<r}v_t\ge A'+1$. So $x=E_{12}$ and $y=A-1-a$. □

Proof of Theorems G and F. Combine (T1)–(T3), Lemma 7 and Lemma 8. For $m\le p^2+p-1$ every $\lambda\le m$ is simple or has two Weyl factors, by (T2). □

Example 9. For $p=q=3$ and $d=4$: Theorem A gives the pieces $X(4,0)$, $X(3,1)^4$, $X(2,2)^6$, $X(1,3)^4$, $X(0,4)$, and Theorem F gives $\mathrm{Syl}_3\,K(H(4,3))\cong(\mathbb{Z}/3)^2\oplus(\mathbb{Z}/3^2)^7\oplus(\mathbb{Z}/3^3)^8\oplus(\mathbb{Z}/3^4)^{17}$, which is the group we computed.

7. The prime 2

For $p=2$ the blocks are the families of [DST], and [DST] computes them for every family and every power of 2. This gives the whole 2-part of $K(H(d,q))$ for every d and every even q.

For a family $\lambda\ge 1$ write $\lambda+1=2^\theta+\sum_{t\in I}2^t$ with $I\subseteq\{0,\dots,\theta-1\}$, $\Theta=2^\theta$, $h_t=\mathrm{next}(t)-t$ (with $\mathrm{next}(t)$ the next element of $I\cup\{\theta\}$), $o_D=\sum_{t\in D}2^t$ for $D\subseteq I$, and $\kappa(a,b)=v_2\binom{a+b}{a}$. Following [DST, §3.4] put

$$X^{(\alpha)}(\lambda,i)=\operatorname{coker}\big(f+2^\alpha(D_\lambda+i)\text{ on }T_{\mathbb{Z}_2}(\lambda)\big)\qquad(\alpha\ge 1,\ i\ge 0),$$

where D_λ is the floor operator $(\lambda-h)/2$. By [DST, Theorem D, Proposition 5.4, Theorem 7.10], which are stated and proved for every $\alpha \geq 1$,

$$X^{(\alpha)}(\lambda, i) \cong \bigoplus_{b=1}^{\theta-1} \big(\mathbb{Z}/2^b\big)^{2^{|I|+\theta-1-b}} \oplus \bigoplus_{D\subseteq I} \mathbb{Z}/2^{\,\theta+\alpha\Theta+\kappa(i-1,\Theta)+\hat{y}_D} \qquad (i \geq 1),$$

where $\hat{y}$ is the integer antitonic fit [DST, §1.2], in the order of increasing o_D, of the clocks

$$x_D = R_{i-1}(D) - R_{i-1+\Theta}(I \setminus D), \qquad R_\mu(E) = \sum_{t\in E} \big(h_t - \alpha 2^t\big) - \kappa(\mu, o_E).$$

For $i = 0$ the summand $D = \emptyset$ is $\mathbb{Z}_2$, and the others are $\mathbb{Z}/2^{\theta+\alpha\Theta+\hat{y}^\sharp_D}$, with $\hat{y}^\sharp$ the fit of $x^\sharp_D = R^\sharp(D) - R^\sharp(I \setminus D)$, $R^\sharp(E) = \sum_{t\in E}(h_t - \alpha 2^t) - \kappa(\Theta - 1, o_E)$, over $D \neq \emptyset$. For $\lambda = 0$, $X^{(\alpha)}(0, i) = \mathbb{Z}/2^{\alpha+v_2(i)}$ ($\mathbb{Z}_2$ if $i = 0$). Let $m_\lambda(n)$ be the multiplicity of $T(\lambda)$ in $V^{\otimes n}$ in characteristic 2 [DST, §1.2, Theorem 3.6].

Theorem H (the prime 2)**.** *Let q be even, $e = v_2(q)$, $m \geq 0$ and $c \geq 0$. Then*

$$\mathbb{Z}_2^{[c=0]} \oplus \mathrm{Syl}_2\, X_q(m, c) \;\cong\; \bigoplus_\lambda X^{(e)}\big(\lambda,\, c + \tfrac{m-\lambda}{2}\big)^{m_\lambda(m)}.$$

Consequently, for every $d \geq 1$,

$$\mathbb{Z}_2 \oplus \mathrm{Syl}_2\, K\big(H(d, q)\big) \;\cong\; \bigoplus_{j=0}^{d} \bigoplus_\lambda X^{(e)}\big(\lambda,\, j + \tfrac{d-j-\lambda}{2}\big)^{\binom{d}{j}(q-2)^j\, m_\lambda(d-j)}.$$

Proof. By (T1) for $p = 2$ [DST, Theorem 3.6], $B_m \otimes \mathbb{Z}_2 = V^{\otimes m} = \bigoplus_\lambda T_{\mathbb{Z}_2}(\lambda)^{m_\lambda(m)}$ as Dist-modules. On a summand $T(\lambda)$, placed at the levels $k = (m-\lambda)/2, \dots, k+\lambda$, a vector of weight h has $H = (m-h)/2 = k + D_\lambda$, so $q(c+H)+U = f + q(c+k+D_\lambda)$. Write $q = 2^e u$ with u odd. The proof of Lemma 2 applies verbatim to any lattice graded by the floor on which f raises the floor by one; it rescales the floors by units and turns $f + 2^e u(D_\lambda + i)$ into $f + 2^e(D_\lambda + i)$. So the cokernel on the summand is $X^{(e)}(\lambda, c+k)$. The second statement follows from Theorem A. □

For $q = 2$ only $j = 0$ survives, $e = 1$, and this is the Main Theorem of [DST]. For $e \geq 1$ arbitrary, the theorem was checked against the Smith forms of $q(c + H) + U$ on B_m in 384 cases ($m \leq 11$, $e \leq 4$, $c \leq 4$) and against the full Laplacians of $H(d, q)$ for $(d, q) = (3, 4), (4, 4), (3, 6), (3, 8), (2, 12), (2, 16), (3, 10)$. Two controls fail: using $\alpha = 1$ instead of $\alpha = e$ changes the answer in all 112 cases with $e \geq 2$ tried, and omitting the antitonic fit changes it in 16 of 128.

8. The remaining blocks: a conjecture

For a prime p, $\lambda \geq 0$, $i \geq 0$ and $e \geq 1$ put

$$Y_p(\lambda, i, e) = \operatorname{coker}\big(f + p^e(D_\lambda + i) \text{ on } T_{\mathbb{Z}_p}(\lambda)\big),$$

with $D_\lambda = (\lambda - h)/2$ the floor operator, as in Section 7. By (T1), and by Lemma 2 applied as in the proof of Theorem H, for every m and c

$$\mathbb{Z}_p^{[c=0]} \oplus \mathrm{Syl}_p\, X_q(m, c) \;\cong\; \bigoplus_\lambda Y_p\big(\lambda,\, c + \tfrac{m-\lambda}{2},\, v_p(q)\big)^{n_\lambda(m)},$$

where $n_\lambda(m)$ is the multiplicity of $T(\lambda)$ in $V^{\otimes m}$. So everything comes down to the groups $Y_p(\lambda, i, e)$. Theorem G computes them for $\lambda \leq p^2 + p - 1$, and Theorem H for $p = 2$.

Two things are known about $Y_p(\lambda, i, e)$ for every λ. Write $\lambda + 1 = \sum_{t=0}^{k} d_t p^t$ with $0 \leq d_t \leq p - 1$ and $d_k \neq 0$, and put $I(\lambda) = \{t < k : d_t \neq 0\}$.

- *The number of cyclic summands* (the free summand included when $i = 0$) is the number of Jordan blocks of f on $T(\lambda)$ over $\mathbb{F}_p$, because $p^e(D_\lambda + i) \equiv 0$. For $\lambda \geq p - 1$ the module $T(\lambda)$ is projective over the first Frobenius kernel [Jan03, Part II, Chapter E], so f acts freely over $\mathbb{F}_p[f]/(f^p)$ and the number is $\dim T(\lambda)/p$.

- *The number of Weyl factors* of $T(\lambda)$ is $2^{|I(\lambda)|}$. Indeed, by Donkin's theorem $T(p-1+r'+pb) \cong T(p-1+r') \otimes T(b)^{[1]}$ $(0 \le r' \le p-1)$, and $\chi(p-1)\cdot\chi(n)(z^p) = \chi(pn+p-1)$, while for $r' \ge 1$ the character of $T(p-1+r')$ is $\chi(p-1)(z^{r'} + z^{-r'})$; so each Weyl factor $\Delta(n)$ of $T(b)$ gives one Weyl factor of $T(\lambda)$ if $r' = 0$ and two, $\Delta(pn+p-1\pm r')$, if $r' \ge 1$. Since $\lambda + 1 = r' + p(b+1)$, this is the recursion $|I(\lambda)| = [r' \ne 0] + |I(b)|$, starting from $T(\lambda) = \Delta(\lambda)$ for $\lambda \le p-2$.

Conjecture I. *Let p be a prime, $\lambda \ge 0$, $i \ge 0$ and $e \ge 1$, and let k and $I(\lambda)$ be as above. Then:*

(i) *exactly $2^{|I(\lambda)|}$ of the cyclic summands of $Y_p(\lambda, i, e)$ have order larger than p^k (the free summand, when $i = 0$, counting as one of them), one for each Weyl factor of $T(\lambda)$;*
(ii) *the others have order at most p^k, and together they form a group that does not depend on i or on e.*

The conjecture holds whenever the groups are known. For $\lambda \le p^2 + p - 1$ it follows from Theorem G: there $k \le 2$, the small summands are $(\mathbb{Z}/p)^a$ or $(\mathbb{Z}/p)^{2a}$, and the large ones are the $\mathbb{Z}/p^x$, $\mathbb{Z}/p^y$ or $\mathbb{Z}/p^{A-a}$ of the block; since every $v_t \ge 1$, one has $A' \ge \lambda^- + 1$, $\alpha \ge r$ and $A - \alpha \ge \lambda + 1 - r$, which gives $E_{11}, E_{12}, E_{22} \ge k+1$ and $A - a \ge k+1$. For $p = 2$ it follows from Theorem H: the large summands are the "big clocks" indexed by $D \subseteq I$, and the small ones are $\bigoplus_{b<\theta} (\mathbb{Z}/2^b)^{2^{|I|+\theta-1-b}}$. We checked both statements on the formulas: 9045 groups for $p = 2$ ($\lambda \le 200$, $e \le 5$, $i \le 8$) and 4704 for $p \in \{3, 5, 7\}$ ($\lambda \le p^2 + p - 1$), with no exception; and the count of Weyl factors for $p \le 5$, $\lambda \le 120$.

The first block outside both cases is $\lambda = 12$ for $p = 3$ ($\lambda + 1 = 111_3$, four Weyl factors). Here $\dim T(12) = 36$, $k = 2$ and $I = \{0, 1\}$, so the conjecture predicts $(\mathbb{Z}/3)^8$ and four summands of order at least 27. The block occurs once in $V^{\otimes 12}$, at the top, and every other block of $V^{\otimes 12}$ has $\lambda \le 10$ and is given by Theorem G. So we computed the Smith form of $q(c + H) + U$ on B_{12} for $q \in \{3, 6\}$ and $0 \le c \le 3$, and removed the blocks $\lambda \le 10$. We stated the conjecture after seeing the cell $(q, c) = (3, 0)$ alone, and sealed a prediction for the other seven (Section 9). The block is:

c	block $\lambda = 12$, for $q = 3$ and for $q = 6$
0	$\mathbb{Z}_3 \oplus (\mathbb{Z}/3)^8 \oplus \mathbb{Z}/3^7 \oplus \mathbb{Z}/3^{10} \oplus \mathbb{Z}/3^{12}$
1, 2	$(\mathbb{Z}/3)^8 \oplus (\mathbb{Z}/3^9)^2 \oplus (\mathbb{Z}/3^{13})^2$
3	$(\mathbb{Z}/3)^8 \oplus \mathbb{Z}/3^8 \oplus \mathbb{Z}/3^{10} \oplus \mathbb{Z}/3^{12} \oplus \mathbb{Z}/3^{14}$

All seven predictions hold. The large summands depend on c; we have no formula for them. For two Weyl factors their exponents are the numbers x, y of Theorem G, and for $p = 2$ they are the big clocks of [DST]; we expect a common form.

9. Verification

All computations were exact. The Smith forms over $\mathbb{Z}_{(p)}$ were computed modulo p^M with $p^M < 2^{31}$, by unit pivots and Schur complements, level by level in the p-adic valuation; factors of valuation at least M were then determined separately, by the matrix $p^E M^{-1}$ of Lemma 4, whose elementary divisors are p^{E-a} for the divisors p^a of M. The scripts, the sealed prediction files and the logs are at `https://github.com/tretoef-estrella/coloured-sand-theorem` and in the Zenodo record doi:10.5281/zenodo.23327608 (all versions).

(1) Theorem A: the Sylow p-subgroups for $p \in \{2, 3, 5, 7\}$ computed directly from the Laplacian and from the pieces agree in all 90 comparisons (27 cells, $q \in \{3, \dots, 10, 12\}$, $q^d \le 729$, including $p \mid q$ and $p \le d$); adding one extra copy of a piece changes the answer in 49 of them. Independently, the cells $(d, q) = (3, 7)$ and $(2, 11)$ were recomputed with Sage from the full Laplacian.
(2) Theorem B: found first as a conjecture from the cells with $d \le 4$; before the proof, ten predictions for $(d, p) = (2, 11), (2, 13), (2, 17), (2, 19), (2, 23), (3, 17), (3, 19), (3, 23), (3, 29), (4, 13)$ were written to a file (MD5 `d5c36be3736079cbc2349f469751c627`) and then computed: ten agreements.

(3) Theorem C: 143 cells, for $q \in \{3, \dots, 10, 12\}$ and $p \in \{2, 3, 5, 7\}$, both with $p \mid q$ (including $p = 2$) and with $p \nmid q$, all in agreement; the guess $ed + v_p(d)$ fails in one of them and the guess ed in 26.

(4) Theorem D: the rank formula for $W_{s,s+1}$ checked directly in 264 cases ($p \le 7$, $m \le 11$); the value of N agrees with the computed groups in 23 cells with $p \le d$.

(5) Theorem F, before its proof (version 1): found from 88 pieces with $p = 3, 5$. Then 30 pieces with $p = 7$, $7 \le m \le 11$, $1 \le c \le 6$ were predicted before being computed (MD5 of the file `1f5cc12e88f12e375022658da3eea6ce`): 30 agreements. Then 14 full groups $\mathrm{Syl}_p K(H(d, q))$ with composite $q \in \{6, 9, 10, 12, 15, 20, 25, 27\}$ and $p \le d \le 2p-2$ (MD5 `341a2c489aef1c30349bae079ddcce81`): 14 agreements. It also agrees with the 10 full groups computed for $q = p \in \{3, 5, 7\}$ and $p \le d \le 2p - 2$.

(6) Lemmas 7 and 8, by machine:
- the lattices of Lemma 7 are stable under every $e^{(t)}$ and $f^{(t)}$ for $p \in \{3, 5, 7\}$ and every $\lambda \in [p, 3p - 2]$ (27 lattices; 2200 checks for $\lambda \le 2p - 2$);
- the cokernel of $q(c + H) + U$ on the model equals the formula of Lemma 8 in 292 random cases with $\lambda \le 2p - 2$ ($p \le 7$), in 180 cases with $\lambda \le p^2 - 2$ ($p = 3, 5$; $e \in \{1, 2\}$), and in 60 cases with $c = 0$;
- removing the glue (the split lattice) changes the answer in 293 of 293 cases, and the formula of Theorem F applied beyond $2p - 2$ fails in 12 of 18 pieces.

(7) Theorem G against direct computation (version 2): the predicted pieces equal the Smith forms of $q(c + H) + U$ computed on B_m in 154 cases ($p = 3$, $m \le 7$; $p = 5$, $m \le 10$; $e \in \{1, 2\}$; $0 \le c \le 6$) and in 5 cases with $p = 3$, $m = 8 = 3p - 1$. The full groups for $p = q = 3$ and $d = 5, \dots, 8$ computed in version 1 from the full Laplacian (Section 10) are reproduced exactly.

(8) Corollary 6: $\dim_{\mathbb{F}_p} \operatorname{coker} U$ on B_m computed by ranks agrees with the count of Jordan blocks in 37 cases ($p \in \{3, 5, 7\}$, $m \le 12$); the characteristic-0 count $\binom{m}{\lfloor m/2 \rfloor}$ differs in 7 of them.

(9) Theorem G up to $m \le p^2 + p - 1$ (version 3):
- the formula of Lemma 8 against the Smith form on the model lattice of Lemma 7, for $p \in \{5, 7\}$ and every $\lambda \le p^2 - 1$, $c = k = 0$ included: 216 of 216; the split lattice differs in 52 of 52; the lattices of Lemma 7 are stable under every $e^{(t)}$ and $f^{(t)}$ for every $\lambda \le p^2 - 1$ with two Weyl factors, $p \in \{5, 7\}$ (52 lattices);
- a first sealed prediction (MD5 `9ef6b4fec30a93192b0de995519d75f3`), written after the formula was derived and before the computation: 36 pieces ($p = 3$, $m \in \{9, 10\}$, $q \in \{12, 15, 21\}$, $5 \le c \le 8$; $p = 2$, $m = 4$, $q \in \{6, 8, 10\}$, $5 \le c \le 8$), the four full groups $\mathrm{Syl}_3 K(H(10, 3))$, $\mathrm{Syl}_3 K(H(9, 6))$, $\mathrm{Syl}_2 K(H(4, 6))$, $\mathrm{Syl}_2 K(H(4, 10))$, and $\mathrm{Syl}_2 K(H(4, 6))$ again from its full Laplacian: 41 agreements;
- a second one (MD5 `a092c60102cfca163c8449b60680b1e1`): 18 pieces ($p = 3$, $m = 11$, $q \in \{12, 15\}$, $4 \le c \le 6$; $p = 2$, $m = 5$, $q \in \{6, 8, 10\}$, $4 \le c \le 7$), the full groups $\mathrm{Syl}_3 K(H(11, 3))$, $\mathrm{Syl}_2 K(H(5, 4))$, $\mathrm{Syl}_2 K(H(5, 6))$, and $\mathrm{Syl}_2 K(H(5, 4))$ from its full Laplacian: 22 agreements;
- the row $d = 9$ of Section 10, which version 2 did not explain, is reproduced exactly.

(10) Theorem H: 384 pieces against Smith forms on B_m ($m \le 11$, $e \le 4$, $c \le 4$) and seven full Laplacians, with the two controls described after the theorem.

(11) Conjecture I: on the formulas of Theorems G and H, 9045 groups for $p = 2$ ($\lambda \le 200$, $e \le 5$, $i \le 8$) and 4704 for $p \in \{3, 5, 7\}$, with no exception; the number of summands equals $\dim T(\lambda)/p$ in 286 of 286 cases; the count $2^{|I(\lambda)|}$ of Weyl factors from the characters for $p \le 5$, $\lambda \le 120$ (363 cases). For the block $\lambda = 12$, $p = 3$: a third sealed prediction (MD5 `a739d9c18622ab0c5dfc811f4cbd49f7`), written after one cell, for seven cells; seven agreements.

10. Data for $p = q = 3$

For $q = p = 3$ the computed groups, written as $(n_1, n_2, \dots)$ with n_v the number of factors $\mathbb{Z}/3^v$, are:

d	$(n_1, n_2, \dots)$
5	$(12, 10, 16, 53, 5)$
6	$(38, 37, 36, 117, 24, 16, 6)$
7	$(86, 105, 127, 285, 85, 63, 41)$
8	$(250, 211, 286, 1078, 174, 133, 174, 8)$
9	$(942, 412, 624, 3339, 449, 292, 495, 133, 84, 37, 9)$
10	$(3314, 1388, 1768, 7310, 1690, 1364, 1388, 1392, 316, 190, 74)$
11	$(11035, 4345, 5127, 11339, 10099, 4994, 4432, 7162, 704, 419, 416)$

Theorem G gives every row exactly. In version 2 the row $d = 9$ was outside the range: its piece $X_3(9, 0)$ contains the block $\lambda = 9$, which has $a = p - 1$, and the version-2 formula, with glue $1/p$, gave $(942, 414, 623, 3339, 448, 293, 495, 134, 84, 36, 9)$. With the glue $1/p^2$ of Lemma 7 and the minimum in the definition of $G_c(m, \lambda)$ the row comes out right. The rows $d = 10$ and $d = 11$ were computed piece by piece and assembled by Theorem A; both were predicted before they were computed (the first two sealed predictions of item 9 of Section 9 contain them).

Provenance

The results were found on the night of 12–13 October 2026 by an AI system, Claude (Anthropic), working for the author: it chose the problem after a search of the literature, wrote the computations, found the formulas, sealed and checked the predictions, and found the proofs; it also found, when checking the literature before publication, that Theorem A is [JNR03, Theorem 13]. Version 2 was written later the same night, also by Claude: the Glue Conjecture of version 1 was proved and extended (Theorems F, G) by writing down the tilting lattices, and the closed form of N (Corollary 6) was proved. Version 3 was written later the same night, also by Claude: Theorem G was extended to $m \le p^2 + p - 1$ and every prime by computing the glue of the blocks with $a = p - 1$ (denominator p^2), Theorem H was obtained by reading the results of [DST], which are stated for every power of 2, at $2^{v_2(q)}$, and Conjecture I was stated from one computed cell. The author takes responsibility for the paper. To the best of our knowledge, after a search of the works citing [JNR03] and of arXiv, Theorems B, C (for $e \ge 1$), D, E, F, G and H, Corollary 6 and Conjecture I are new; version 1 also claimed the case $e = 0$ of Theorem C, which follows from [JNR03, Theorem 2]. Theorem A is due to Jacobson, Niedermaier and Reiner. The tilting theory used in Sections 6 and 8, including the count of Weyl factors, is classical [Don93, Jan03], and so is the decomposition of tensor powers in the proof of Corollary 6.

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